

# TP1 - Indicadores de Comercio Internacional: Marruecos

VCR (Balassa), IIC (Yeats), ICC (Michaely) - Economía Internacional, FCE-UBA

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## Introducción

Este documento presenta el cálculo de los principales indicadores de comercio internacional (VCR, IIC, ICC) para Marruecos, país asignado en el TP1 de Economía Internacional. Se utilizan bases de UN Comtrade/WITS bajo nomenclatura CUCI Rev. 3 a 3 dígitos, para el período 2021-2025.

```
library(tidyverse)
library(haven)
library(ggrepel)
library(RColorBrewer)
library(scales)
```

## Configuración de rutas y estilo

```
ruta_datos <- "bases de datos/"
ruta_graficos <- "graficos/01/"
```

```
paleta_socios <- c(
  "ESP" = "#AA151B", "FRA" = "#0055A4", "DEU" = "#FFCE00",
  "USA" = "#3C3B6E", "BRA" = "#009739"
)
```

```
nombres_socio <- c(ESP = "España", FRA = "Francia", DEU = "Alemania",
  USA = "EE.UU.", BRA = "Brasil")
```

```
theme_tp1 <- function(base_size = 13) {
  theme_minimal(base_size = base_size, base_family = "sans") +
    theme(
      plot.title = element_text(face = "bold", size = base_size + 2, color = "gray10"),
      plot.subtitle = element_text(color = "gray35", size = base_size - 1.5, margin = margin(b = 10)),
      plot.caption = element_text(color = "gray55", size = base_size - 5, hjust = 0, margin = margin(t = 10)),
      axis.title = element_text(color = "gray25", size = base_size - 1),
      axis.text = element_text(color = "gray40"),
      legend.title = element_text(face = "bold", size = base_size - 1),
      legend.text = element_text(size = base_size - 2),
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(color = "gray92", linewidth = 0.3),
      strip.text = element_text(face = "bold", size = base_size - 1)
    )
}
```

## Carga y limpieza de datos

Se utilizan 3 bases: Marruecos como reporter contra 77 socios + el mundo, los 5 socios finales como reporters contra el mundo (para el ICC), y el agregado mundial (para el denominador del VCR).

```

limpiar_wits <- function(path, incluir_desc = TRUE) {
  base <- read_dta(path)
  names(base) <- tolower(names(base))
  base <- base %>%
    mutate(
      cuci = as.character(as_factor(productcode)),
      r     = as.character(as_factor(reporteriso3)),
      p     = as.character(as_factor(partneriso3)),
      flow  = as.character(as_factor(tradeflowname))
    ) %>%
    rename(value = tradevaluein1000usd)
  if (incluir_desc) {
    base <- base %>%
      mutate(cuci_desc = as.character(as_factor(productdescription))) %>%
      select(r, p, flow, cuci, cuci_desc, value, year)
  } else {
    base <- base %>% select(r, p, flow, cuci, value, year)
  }
  base
}

comtrade_mar <- limpiar_wits(paste0(ruta_datos, "MAR_ALLPARTNERS_WLD.dta"))
comtrade_socios <- limpiar_wits(paste0(ruta_datos, "SOCIOS_WLD.dta"))
comtrade_wld <- limpiar_wits(paste0(ruta_datos, "WLD_WLD.dta"), incluir_desc = FALSE)

mar_exp <- comtrade_mar |>
  filter(flow == "Export") |>
  group_by(year, p) |>
  mutate(total_expo = sum(value, na.rm = TRUE)) |>
  ungroup() |>
  mutate(share = value / total_expo)

mar_imp <- comtrade_mar |>
  filter(flow == "Import") |>
  group_by(year, p) |>
  mutate(total_impo = sum(value, na.rm = TRUE)) |>
  ungroup() |>
  mutate(share = value / total_impo)

wld_exp <- comtrade_wld |>
  filter(flow == "Export", r == "All", p == "All") |>
  group_by(year) |>
  mutate(total_wld = sum(value, na.rm = TRUE)) |>
  ungroup() |>
  mutate(share_wld = value / total_wld) |>
  select(year, cuci, share_wld)

```

## Perfil de comercio exterior

### Principales socios comerciales

```

top_socios <- mar_exp |>
  filter(p != "WLD", year == 2024) |>
  group_by(p) |>

```

```
summarise(total = first(total_expo)) |>
  arrange(desc(total)) |>
  slice_max(total, n = 15)
```

```
knitr::kable(top_socios, caption = "Top 15 socios comerciales de Marruecos por exportaciones (2024)")
```

Table 1: Top 15 socios comerciales de Marruecos por exportaciones (2024)

| p   | total      |
|-----|------------|
| ESP | 18209308.1 |
| FRA | 15837620.6 |
| DEU | 4401897.0  |
| ITA | 4229035.2  |
| GBR | 3533492.1  |
| IND | 2692825.2  |
| BRA | 2624626.2  |
| USA | 2420911.5  |
| NLD | 1326233.0  |
| PRT | 1298940.0  |
| PAK | 1229210.8  |
| BEL | 1144236.0  |
| ROM | 1038297.0  |
| AUS | 879862.6   |
| HKG | 733122.6   |

### Destino de los fertilizantes (hallazgo clave del TP)

```
destino_fertilizantes <- comtrade_mar |>
  filter(flow == "Export", year == 2024, cucu %in% c("272", "562"), p != "WLD") |>
  group_by(p, cucu_desc) |>
  summarise(total = sum(value, na.rm = TRUE), .groups = "drop") |>
  arrange(desc(total)) |>
  slice_max(total, n = 15)
```

```
knitr::kable(destino_fertilizantes, caption = "Principales destinos de fertilizantes crudos y manufacturados (2024)")
```

Table 2: Principales destinos de fertilizantes crudos y manufacturados (2024)

| p   | cucu_desc                | total     |
|-----|--------------------------|-----------|
| BRA | Manufactured fertilizers | 2404754.1 |
| IND | Manufactured fertilizers | 1630921.5 |
| AUS | Manufactured fertilizers | 825005.7  |
| ARG | Manufactured fertilizers | 634934.9  |
| USA | Manufactured fertilizers | 545561.4  |
| CAN | Manufactured fertilizers | 369300.3  |
| FRA | Manufactured fertilizers | 249751.6  |
| ESP | Manufactured fertilizers | 248112.3  |
| IND | Fertilizers crude        | 217187.6  |
| MEX | Fertilizers crude        | 173920.8  |

| p   | cuci_desc                | total    |
|-----|--------------------------|----------|
| ROM | Manufactured fertilizers | 139539.4 |
| NGA | Manufactured fertilizers | 130819.1 |
| GBR | Manufactured fertilizers | 129959.0 |
| BEL | Manufactured fertilizers | 126390.1 |
| ITA | Manufactured fertilizers | 116558.3 |

### Principales sectores a 2 dígitos

```
comtrade_2dig <- limpiar_wits(paste0(ruta_datos, "MAR_2DIG_WLD.dta"))

lookup_2dig <- comtrade_2dig |>
  filter(r == "MAR", p == "WLD") |>
  distinct(cuci, cucu_desc) |>
  rename(cuci_2d = cucu, desc_2d = cucu_desc)

sectores_2dig_exp <- mar_exp |>
  filter(p == "WLD", year == 2024) |>
  mutate(cuci_2d = substr(cuci, 1, 2)) |>
  group_by(cuci_2d) |>
  summarise(total = sum(value, na.rm = TRUE)) |>
  arrange(desc(total)) |>
  slice_max(total, n = 5) |>
  left_join(lookup_2dig, by = "cuci_2d") |>
  relocate(desc_2d, .after = cucu_2d)

knitr::kable(sectores_2dig_exp, caption = "Principales sectores exportadores a 2 dígitos (2024)")
```

Table 3: Principales sectores exportadores a 2 dígitos (2024)

| cuci_2d | desc_2d                  | total    |
|---------|--------------------------|----------|
| 78      | Road vehicles            | 16395173 |
| 77      | Electrical equipment     | 15803618 |
| 56      | Manufactured fertilizers | 12635121 |
| 84      | Apparel/clothing/access  | 7786711  |
| 05      | Vegetables and fruit     | 4639179  |

```
sectores_2dig_imp <- mar_imp |>
  filter(p == "WLD", year == 2024) |>
  mutate(cuci_2d = substr(cuci, 1, 2)) |>
  group_by(cuci_2d) |>
  summarise(total = sum(value, na.rm = TRUE)) |>
  arrange(desc(total)) |>
  slice_max(total, n = 5) |>
  left_join(lookup_2dig, by = "cuci_2d") |>
  relocate(desc_2d, .after = cucu_2d)

knitr::kable(sectores_2dig_imp, caption = "Principales sectores importadores a 2 dígitos (2024)")
```

Table 4: Principales sectores importadores a 2 dígitos (2024)

| cuci_2d | desc_2d                  | total    |
|---------|--------------------------|----------|
| 77      | Electrical equipment     | 10523216 |
| 78      | Road vehicles            | 10490036 |
| 33      | Petroleum and products   | 8123339  |
| 65      | Textile yarn/fabric/art. | 6647521  |
| 71      | Power generating equipmt | 3855136  |

```

detalle_composicion <- mar_exp |>
  filter(p == "WLD", year == 2024, substr(cuci, 1, 2) %in% sectores_2dig_exp$cuci_2d) |>
  mutate(cuci_2d = substr(cuci, 1, 2)) |>
  left_join(lookup_2dig, by = "cuci_2d") |>
  group_by(cuci_2d) |>
  mutate(
    rank_en_division = rank(-value, ties.method = "first"),
    pct_division = value / sum(value) * 100
  ) |>
  ungroup() |>
  mutate(
    categoria_rank = case_when(
      rank_en_division == 1 ~ "Producto principal",
      rank_en_division == 2 ~ "2° producto",
      rank_en_division == 3 ~ "3° producto",
      TRUE ~ "Resto"
    ) |> factor(levels = c("Producto principal", "2° producto", "3° producto", "Resto"))
  )

# Orden de apilado explícito: Resto abajo (cerca de 0), Producto principal arriba (extremo)
orden_apilado <- c("Resto", "3° producto", "2° producto", "Producto principal")
detalle_composicion <- detalle_composicion |>
  mutate(categoria_rank = factor(categoria_rank, levels = orden_apilado))

# Calculamos manualmente el acumulado por división, en el orden de apilado
detalle_apilado <- detalle_composicion |>
  arrange(desc_2d, categoria_rank) |>
  group_by(desc_2d) |>
  mutate(
    ymax = cumsum(value),
    ymin = ymax - value,
    y_centro = (ymin + ymax) / 2
  ) |>
  ungroup()

# Etiqueta del producto principal: centrada en su segmento real
etiquetas_top <- detalle_apilado |>
  filter(rank_en_division == 1) |>
  mutate(label_completo = paste0(round(pct_division), "%"))

# Totales por división, para el label del extremo derecho
totales_division <- detalle_apilado |>
  group_by(desc_2d) |>
  summarise(total = max(ymax), .groups = "drop")

```

```

paleta_composicion <- c(
  "Producto principal" = "#1b4965",
  "2° producto"       = "#5fa8d3",
  "3° producto"       = "#bee9e8",
  "Resto"             = "#e8e8e8"
)

g_composicion <- ggplot(detalle_apilado, aes(x = reorder(desc_2d, ymax, max))) +
  geom_rect(aes(xmin = as.numeric(reorder(desc_2d, ymax, max)) - 0.38,
               xmax = as.numeric(reorder(desc_2d, ymax, max)) + 0.38,
               ymin = ymin, ymax = ymax, fill = categoria_rank),
            color = "white", linewidth = 0.6) +
  geom_text(
    data = etiquetas_top, aes(y = y_centro, label = label_completo),
    color = "white", fontface = "bold", size = 4
  ) +
  geom_text(
    data = totales_division, aes(x = desc_2d, y = total, label = scales::label_number(scale = 1/1000, s
    hjust = -0.15, size = 3.8, fontface = "bold", color = "gray25", inherit.aes = FALSE
  ) +
  coord_flip(clip = "off") +
  scale_fill_manual(values = paleta_composicion) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.14)), labels = NULL) +
  labs(
    title = "¿Qué tan concentrada está cada división exportadora?",
    subtitle = "Marruecos, 2024 · el % indica el peso del producto principal dentro de su división",
    x = NULL, y = NULL, fill = "Posición dentro\nde la división",
    caption = "Total exportado por división indicado al final de cada barra (millones de US$).")
  ) +
  theme_tpi(base_size = 13) +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 12, face = "bold", color = "gray20"),
    panel.grid.major.x = element_blank(),
    panel.grid.major.y = element_blank(),
    legend.position = "top",
    legend.justification = "left",
    plot.title.position = "plot",
    plot.margin = margin(t = 10, r = 45, b = 10, l = 10)
  )

g_composicion

ggsave(paste0(ruta_graficos, "grafico_composicion_2dig.png"), g_composicion,
        width = 10, height = 6.5, dpi = 300, bg = "white")

```

## VCR - Ventajas Comparativas Reveladas (Balassa, 1965)

```

vcr_aux_mar <- mar_exp |>
  filter(p == "WLD") |>
  select(year, cuci, cuci_desc, share_mar = share)

vcr_mar <- vcr_aux_mar |>

```

¿Qué tan concentrada está cada división exportadora?

Marruecos, 2024 · el % indica el peso del producto principal dentro de su división

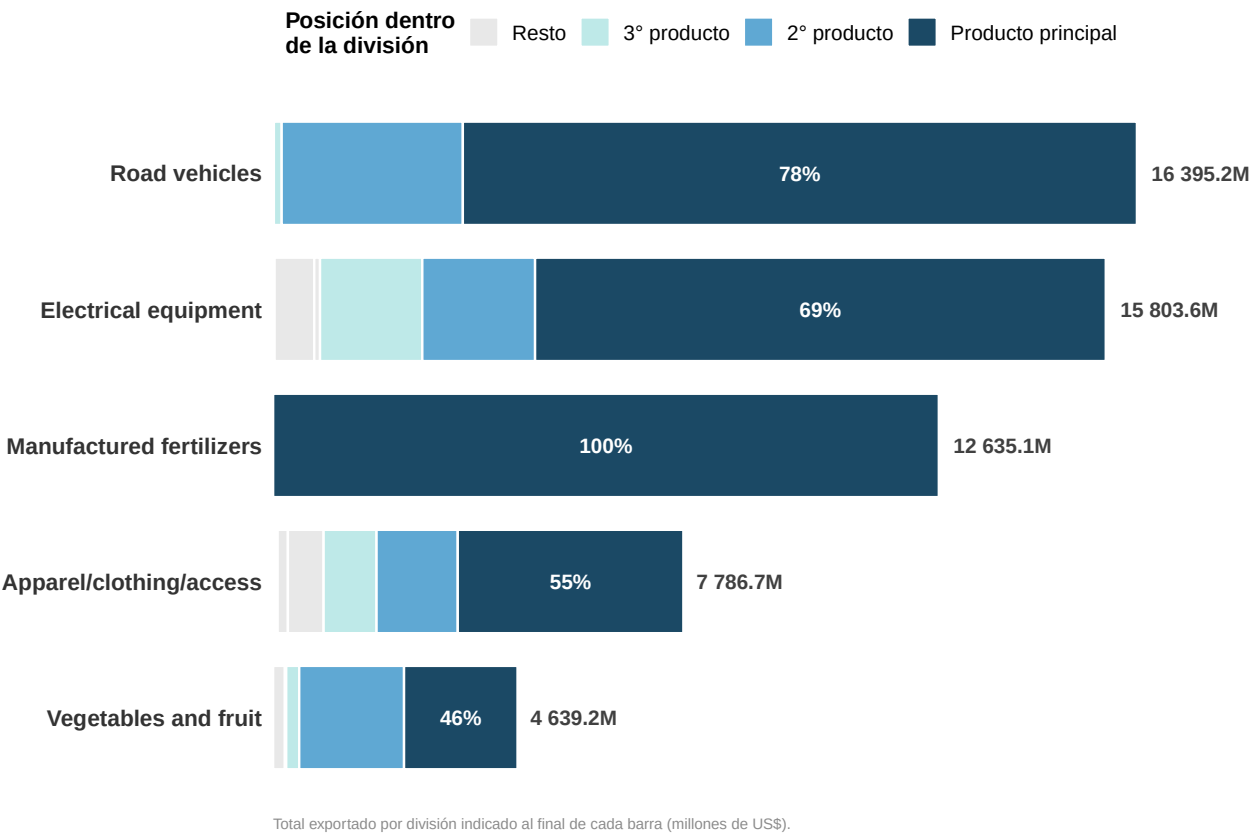


Figure 1: ¿Qué tan concentrada está cada división exportadora?

```

left_join(wld_exp, by = c("year", "cuci")) |>
mutate(vcr = share_mar / share_wld, vcrn = (vcr - 1) / (vcr + 1)) |>
arrange(desc(vcr))

top_ventaja <- vcr_mar |> filter(year == 2024) |> slice_max(vcrn, n = 12)
sectores_top <- top_ventaja$cuci

knitr::kable(top_ventaja |> select(cuci, cuci_desc, vcr, vcrn), digits = 3,
  caption = "Top 12 sectores con mayor VCR/VCRN (2024)")

```

Table 5: Top 12 sectores con mayor VCR/VCRN (2024)

|     | cuci | cuci_desc                | vcr    | vcrn  |
|-----|------|--------------------------|--------|-------|
| 272 |      | Fertilizers crude        | 51.397 | 0.962 |
| 562 |      | Manufactured fertilizers | 51.322 | 0.962 |
| 773 |      | Electrical distrib equip | 16.683 | 0.887 |
| 522 |      | Elements/oxides/hal salt | 11.858 | 0.844 |
| 842 |      | Women/girl clothing wven | 11.816 | 0.844 |
| 036 |      | Crustaceans molluscs etc | 10.484 | 0.826 |
| 037 |      | Fish/shellfish,prep/pres | 9.916  | 0.817 |
| 054 |      | Vegetables,frsh/chld/frz | 6.221  | 0.723 |
| 244 |      | Cork natural/raw/waste   | 6.128  | 0.719 |
| 291 |      | Crude animal mterial nes | 5.293  | 0.682 |
| 411 |      | Animal oil/fat           | 4.783  | 0.654 |
| 792 |      | Aircraft/spacecraft/etc  | 4.756  | 0.653 |

```

variacion_vcrn <- vcr_mar |>
  filter(cuci %in% sectores_top, year %in% c(2021, 2025)) |>
  select(cuci, cuci_desc, year, vcrn) |>
  pivot_wider(names_from = year, values_from = vcrn, names_prefix = "y") |>
  mutate(caida = y2025 - y2021) |>
  arrange(caida)

top_vcrn_2024 <- vcr_mar |> filter(year == 2024) |> slice_max(vcrn, n = 3) |> pull(cuci)
mayor_caida <- variacion_vcrn |> slice_min(caida, n = 2) |> pull(cuci)
sectores_reducidos <- union(top_vcrn_2024, mayor_caida)

g_vcrn <- vcr_mar |>
  filter(cuci %in% sectores_reducidos) |>
  ggplot(aes(x = year, y = vcrn, color = cuci_desc)) +
  geom_line(linewidth = 1.2) + geom_point(size = 2.2) +
  scale_color_brewer(palette = "Dark2") +
  labs(title = "Evolución del VCRN - sectores clave y de mayor caída",
    subtitle = "Marruecos, 2021-2025", x = "Año", y = "VCRN", color = "Producto") +
  theme_tp1()
g_vcrn

ggsave(paste0(ruta_graficos, "grafico_evolucion_vcrn.png"), g_vcrn,
  width = 10, height = 6.5, dpi = 300, bg = "white")

```

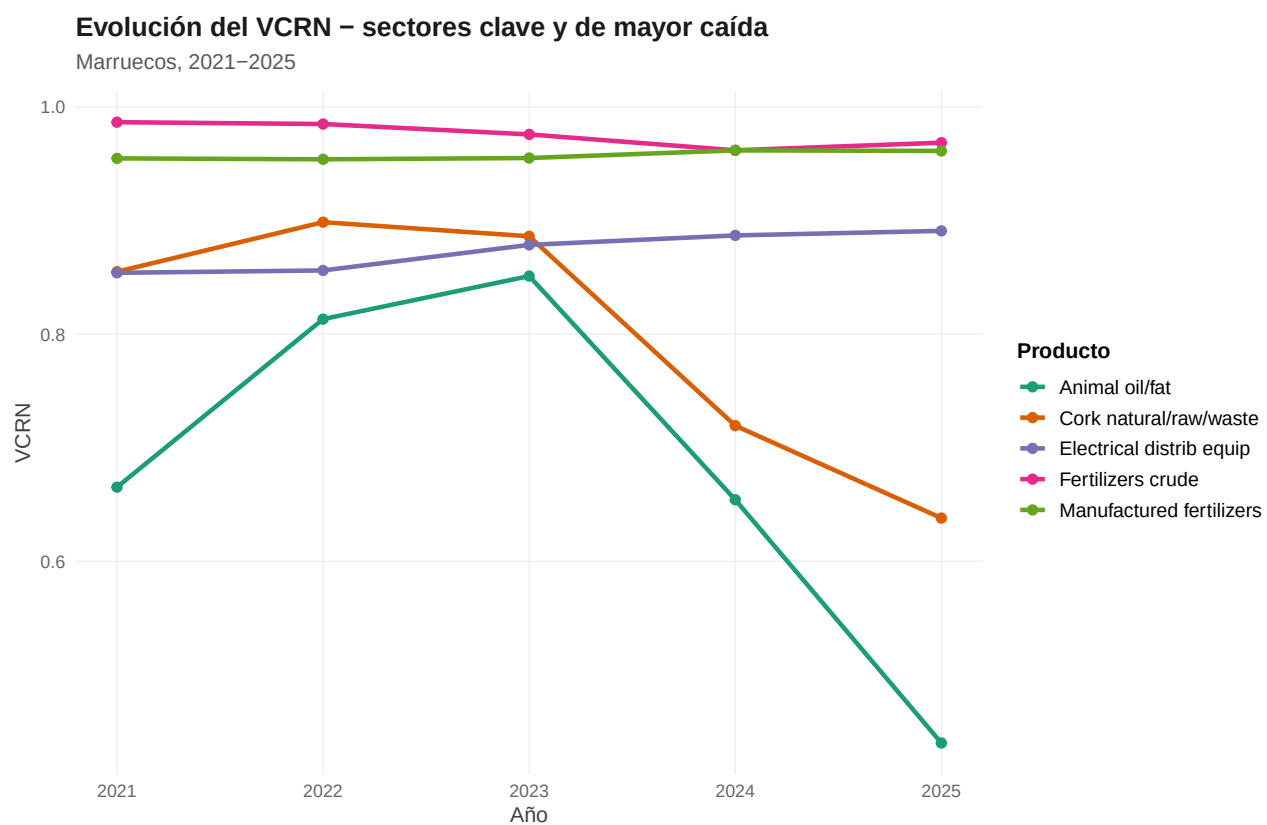


Figure 2: Evolución del VCRN — sectores clave y de mayor caída

## IIC - Índice de Intensidad Comercial (Yeats, 1997)

```
calcular_iic_multi <- function(socio, sectores, anio) {
  mar_soc <- mar_exp |>
    filter(p == socio, year == anio, cuci %in% sectores) |>
    select(cuci, cuci_desc, xijk = value, xij = total_expo)
  mar_wld <- mar_exp |>
    filter(p == "WLD", year == anio, cuci %in% sectores) |>
    select(cuci, xik = value, xi = total_expo)
  mar_soc |>
    left_join(mar_wld, by = "cuci") |>
    mutate(iic = as.numeric((xijk / xik) / (xij / xi)), year = anio) |>
    select(year, cuci, cuci_desc, iic)
}

top_volumen <- mar_exp |>
  filter(p == "WLD", year == 2024) |>
  select(cuci, cuci_desc, value) |>
  arrange(desc(value)) |>
  slice_max(value, n = 12)

socios_finales <- c("ESP", "FRA", "DEU", "USA", "BRA")
sectores_volumen <- top_volumen$cuci

iic_heatmap <- map_dfr(socios_finales, function(s) {
  calcular_iic_multi(s, sectores_volumen, 2024) |> mutate(socio = s)
})

iic_heatmap_completo <- iic_heatmap |>
  tidyr::complete(socio = socios_finales, cuci, fill = list(iic = 0)) |>
  select(-cuci_desc) |>
  left_join(iic_heatmap |> distinct(cuci, cuci_desc), by = "cuci")

iic_heatmap_cat <- iic_heatmap_completo |>
  mutate(
    socio_label = factor(socio, levels = names(paleta_socios), labels = nombres_socio[names(paleta_socios) == socio]),
    categoria = case_when(
      iic == 0 ~ "Sin comercio",
      iic < 0.5 ~ "Muy subrepresentado (<0.5)",
      iic < 1 ~ "Subrepresentado (0.5-1)",
      iic < 2 ~ "Intenso (1-2)",
      TRUE ~ "Muy intenso (>2)"
    ) |> factor(levels = c("Sin comercio", "Muy subrepresentado (<0.5)",
                           "Subrepresentado (0.5-1)", "Intenso (1-2)", "Muy intenso (>2)"))
  )

g_heatmap <- ggplot(iic_heatmap_cat, aes(x = socio_label, y = reorder(cuci_desc, iic), fill = categoria)) +
  geom_tile(color = "gray95", linewidth = 0.8) +
  geom_text(aes(label = ifelse(iic == 0, "-", round(iic, 2))), size = 3, color = "gray15") +
  scale_fill_manual(values = c(
    "Sin comercio" = "gray88", "Muy subrepresentado (<0.5)" = "#fee0d2",
    "Subrepresentado (0.5-1)" = "#fc9272", "Intenso (1-2)" = "#a1d99b", "Muy intenso (>2)" = "#31a354"
  )) +
  labs(title = "IIC de los bienes más exportados por Marruecos, por socio comercial",
```

```

    subtitle = "2024 · bienes ordenados por volumen exportado al mundo",
    x = "Socio comercial", y = NULL, fill = "Intensidad",
    caption = "IIC = 1 indica comercio bilateral proporcional al patrón exportador global de Marruecos",
    theme_tp1(base_size = 12) +
    theme(panel.grid = element_blank(),
          panel.background = element_rect(fill = "white", color = NA),
          plot.background = element_rect(fill = "white", color = NA),
          axis.ticks = element_blank())
g_heatmap

```

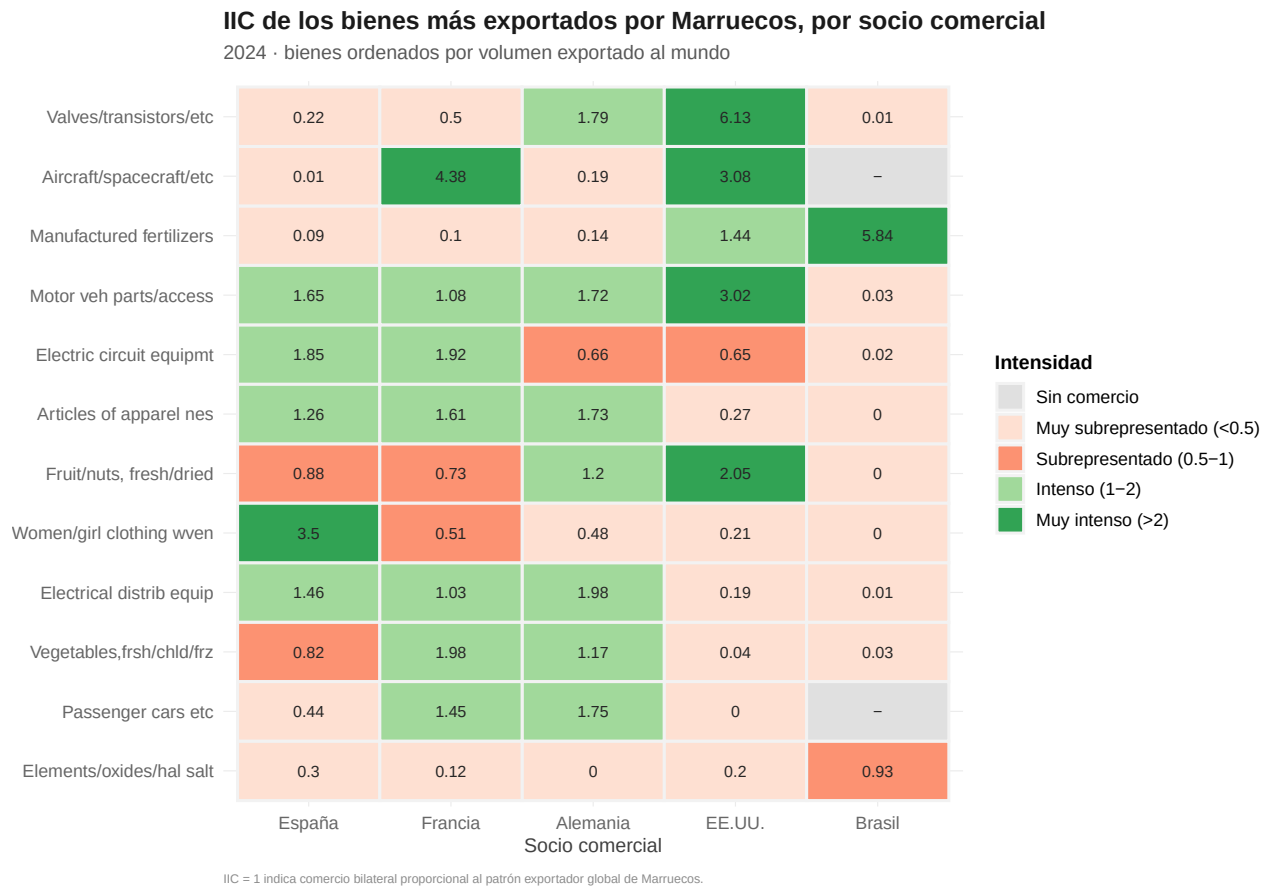


Figure 3: IIC de los bienes más exportados por Marruecos, por socio comercial

```

ggsave(paste0(ruta_graficos, "grafico_heatmap_iic.png"), g_heatmap,
       width = 11, height = 7, dpi = 300, bg = "white")

```

## ICC - Índice de Complementariedad Comercial (Michaely, 1996)

```

calcular_icc <- function(socio, anio) {
  m_mar <- mar_imp |> filter(p == "WLD", year == anio) |> select(cuci, cuc_desc, m = share)
  x_socio <- comtrade_socios |>
    filter(r == socio, p == "WLD", flow == "Export", year == anio) |>
    mutate(x = value / sum(value, na.rm = TRUE)) |>
    select(cuci, x)
  tabla <- full_join(m_mar, x_socio, by = "cuci") |>

```

```

  mutate(m = replace_na(m, 0), x = replace_na(x, 0), dif_abs = abs(m - x))
  icc <- 100 * (1 - sum(tabla$dif_abs, na.rm = TRUE) / 2)
  list(icc = icc, detalle = tabla)
}

candidatos <- c("ESP", "FRA", "DEU", "BRA", "IND", "CHN", "ARE", "GBR", "USA")
ranking_icc <- map_dfr(candidatos, ~tibble(socio = .x, icc = calcular_icc(.x, 2024)$icc)) |>
  arrange(desc(icc))

knitr::kable(ranking_icc, digits = 1, caption = "Ranking de ICC de Marruecos por socio comercial (2024)")

```

Table 6: Ranking de ICC de Marruecos por socio comercial (2024)

| socio | icc  |
|-------|------|
| DEU   | 56.6 |
| USA   | 56.2 |
| ESP   | 54.4 |
| FRA   | 54.3 |
| ARE   | 50.0 |
| IND   | 49.9 |
| CHN   | 48.0 |
| GBR   | 45.0 |
| BRA   | 29.8 |

```

icc_evolucion <- map_dfr(2021:2025, function(a) {
  map_dfr(socios_finales, ~tibble(year = a, socio = .x, icc = calcular_icc(.x, a)$icc))
})

g_icc <- icc_evolucion |>
  filter(socio != "BRA") |>
  ggplot(aes(x = year, y = icc, color = socio)) +
  geom_line(linewidth = 1.1) + geom_point(size = 2) +
  scale_color_manual(values = paleta_socios, labels = nombres_socio) +
  labs(title = "Evolución del ICC de Marruecos por socio comercial",
       subtitle = "España, Francia, Alemania y EE.UU. - 2021-2025",
       x = "Año", y = "ICC", color = "Socio",
       caption = "Se excluye Brasil (ICC estructuralmente bajo) para no distorsionar la escala.") +
  theme_tp1()
g_icc

ggsave(paste0(ruta_graficos, "grafico_evolucion_icc.png"), g_icc,
       width = 10, height = 6.5, dpi = 300, bg = "white")

```

## Gráfico integrador: VCR x IIC x Volumen, por socio

```

armar_cruce <- function(socio, anio = 2024) {
  vcr_mar |>
  filter(year == anio) |>
  select(cuci, cuci_desc, vcrn) |>
  inner_join(calcular_iic_multi(socio, unique(vcr_mar$cuci), anio) |> select(cuci, iic), by = "cuci")
  filter(!is.na(iic), is.finite(iic), iic > 0) |>
  left_join(mar_exp |> filter(p == socio, year == anio) |> select(cuci, value), by = "cuci") |>

```

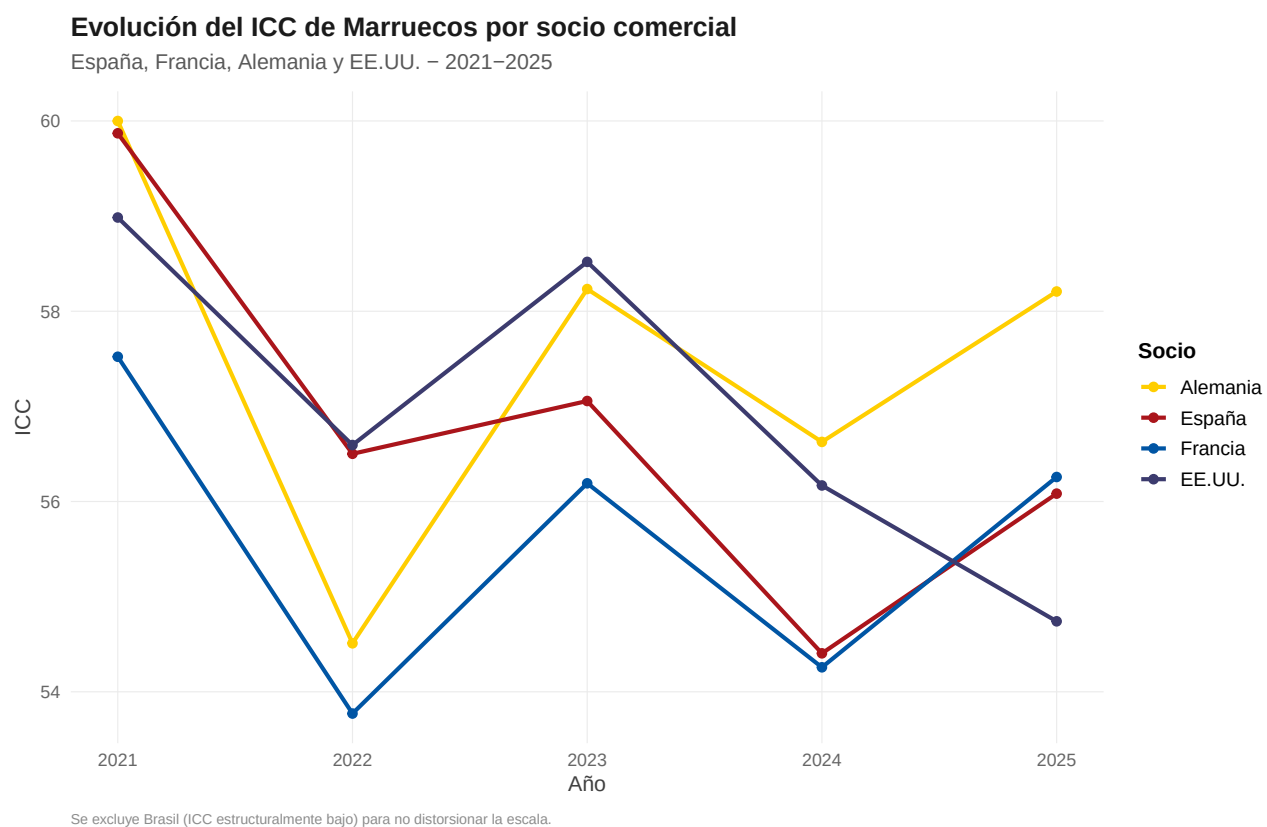


Figure 4: Evolución del ICC de Marruecos por socio comercial

```

    filter(!is.na(value), value > 0) |>
    mutate(socio = socio)
}

destacados_ids <- c("562", "272", "842", "773", "522")

graficar_bubble <- function(socio_code, iic_min = 0.02, n_extra = 3) {
  datos <- armar_cruce(socio_code) |> filter(iic >= iic_min)

  destacados_fijos <- datos |> filter(cuci %in% destacados_ids)
  destacados_extra <- datos |>
    filter(iic > 1, !cuci %in% destacados_ids) |>
    slice_max(value, n = n_extra)
  destacados_s <- bind_rows(destacados_fijos, destacados_extra) |>
    distinct(cuci, .keep_all = TRUE)

  n_dest <- nrow(destacados_s)
  colores_solidos <- RColorBrewer::brewer.pal(max(n_dest, 3), "Set1")[1:n_dest]
  nombres(colores_solidos) <- destacados_s$cuci_desc
  colores_relleno <- scales::alpha(colores_solidos, 0.35)

  rango_valor <- range(sqrt(datos$value))
  destacados_s <- destacados_s |>
    mutate(size_real = scales::rescale(sqrt(value), from = rango_valor, to = c(1, 14)))

  ggplot(datos, aes(x = vcrn, y = iic)) +
    annotate("rect", xmin = 0, xmax = Inf, ymin = 1, ymax = Inf, fill = "steelblue", alpha = 0.06) +
    annotate("rect", xmin = 0, xmax = Inf, ymin = 0, ymax = 1, fill = "firebrick", alpha = 0.06) +
    geom_hline(yintercept = 1, linetype = "dashed", color = "gray40", linewidth = 0.4) +
    geom_vline(xintercept = 0, linetype = "dashed", color = "gray40", linewidth = 0.4) +
    geom_point(aes(size = value), shape = 21, color = "gray50", fill = scales::alpha("gray50", 0.15), size = 21,
    geom_point(data = destacados_s, aes(size = value, color = cuc_i_desc, fill = cuc_i_desc), shape = 21,
    geom_text_repel(
      data = destacados_s, aes(label = cuc_i_desc, color = cuc_i_desc),
      size = 3.3, fontface = "bold", show.legend = FALSE,
      point.size = destacados_s$size_real, box.padding = 1, min.segment.length = 0,
      segment.color = "gray50", force = 10, max.iter = 15000
    ) +
    scale_y_log10(labels = scales::label_number(accuracy = 0.01)) +
    scale_x_continuous(limits = c(-1.05, 1.05), breaks = seq(-1, 1, 0.5)) +
    scale_size_continuous(range = c(1, 14)) +
    scale_color_manual(values = colores_solidos) +
    scale_fill_manual(values = colores_relleno) +
    guides(color = "none", fill = "none", size = "none") +
    labs(
      title = paste("VCR vs. IIC - Marruecos hacia", nombres_socio[socio_code]),
      subtitle = "2024 · el tamaño del punto es el valor exportado",
      x = "VCRN - ventaja comparativa revelada normalizada", y = "IIC (escala log)",
      caption = paste0(
        "Línea punteada horizontal: IIC = 1 · vertical: VCRN = 0\n",
        "Se excluyen productos con IIC < ", iic_min, ". Etiquetas: 5 productos de referencia",
        if (n_extra > 0) paste0(" + hasta ", n_extra, " de mayor volumen con IIC > 1 propios de este socio")
      )
    )
}

```

```

) +
  theme_tp1()
}

umbrales_iic <- c(ESP = 0.02, FRA = 0.02, DEU = 0.001, USA = 0.02, BRA = 0.02)

```

## España

```

p_esp <- graficar_bubble("ESP", iic_min = umbrales_iic[["ESP"]])
p_esp

```

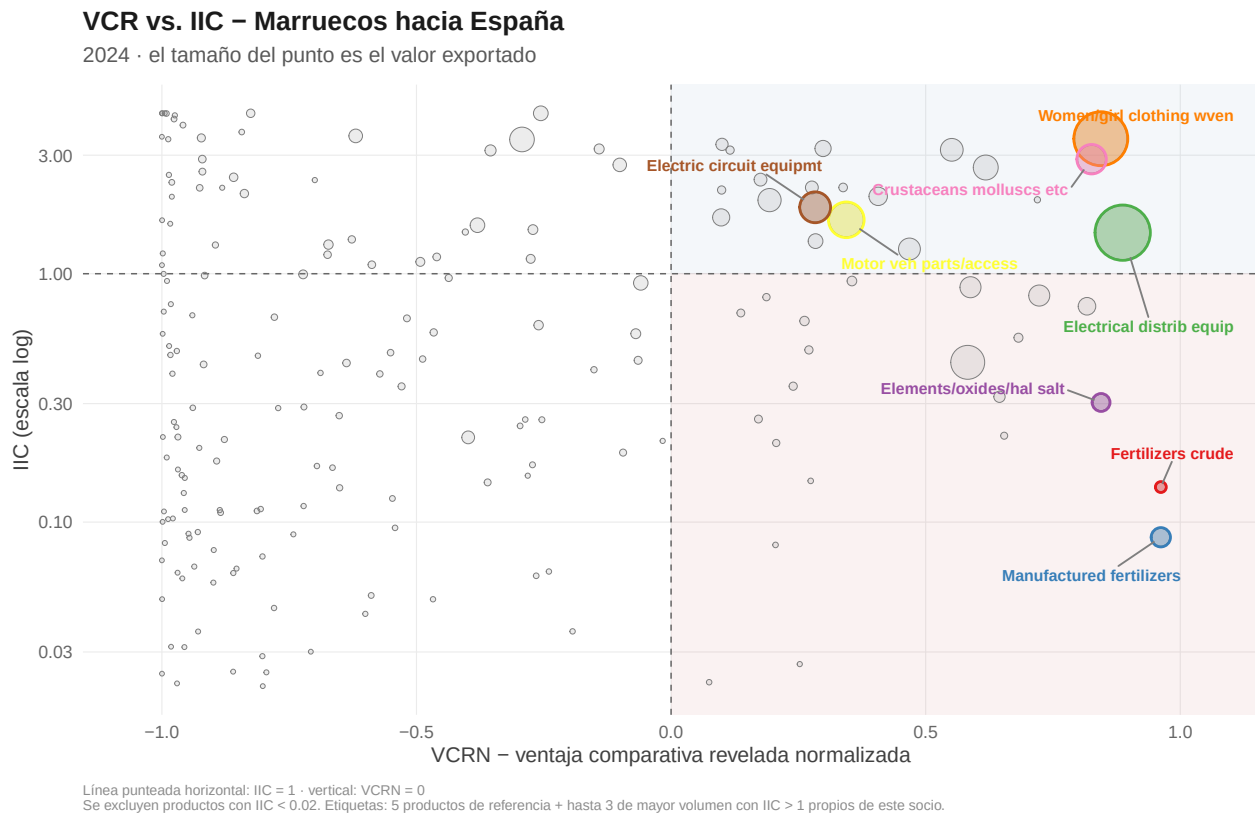


Figure 5: VCR vs. IIC — Marruecos hacia España

```

ggsave(paste0(ruta_graficos, "bubble_ESP.png"), p_esp, width = 10, height = 6.5, dpi = 300, bg = "white")

```

## Francia

```

p_fra <- graficar_bubble("FRA", iic_min = umbrales_iic[["FRA"]])
p_fra

```

```

ggsave(paste0(ruta_graficos, "bubble_FRA.png"), p_fra, width = 10, height = 6.5, dpi = 300, bg = "white")

```

## Alemania

```

p_deu <- graficar_bubble("DEU", iic_min = umbrales_iic[["DEU"]])
p_deu

```

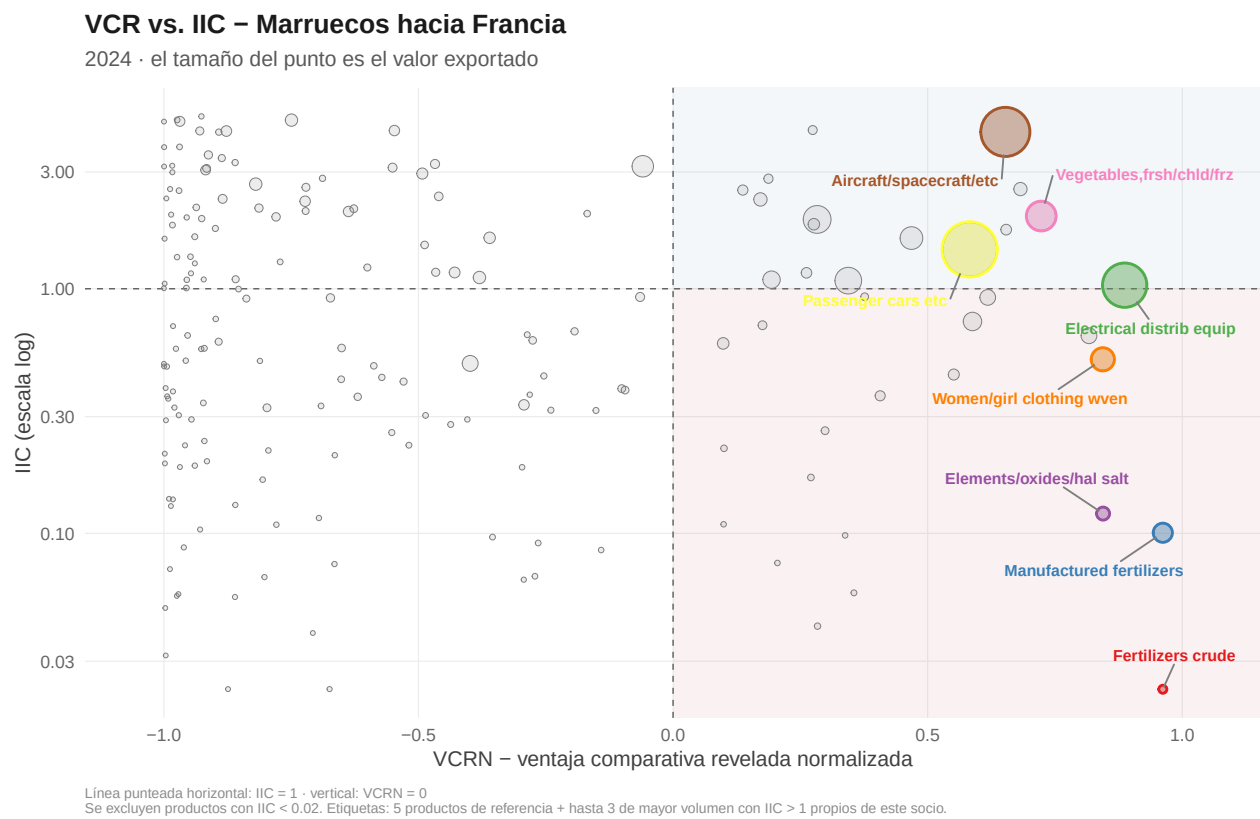


Figure 6: VCR vs. IIC — Marruecos hacia Francia

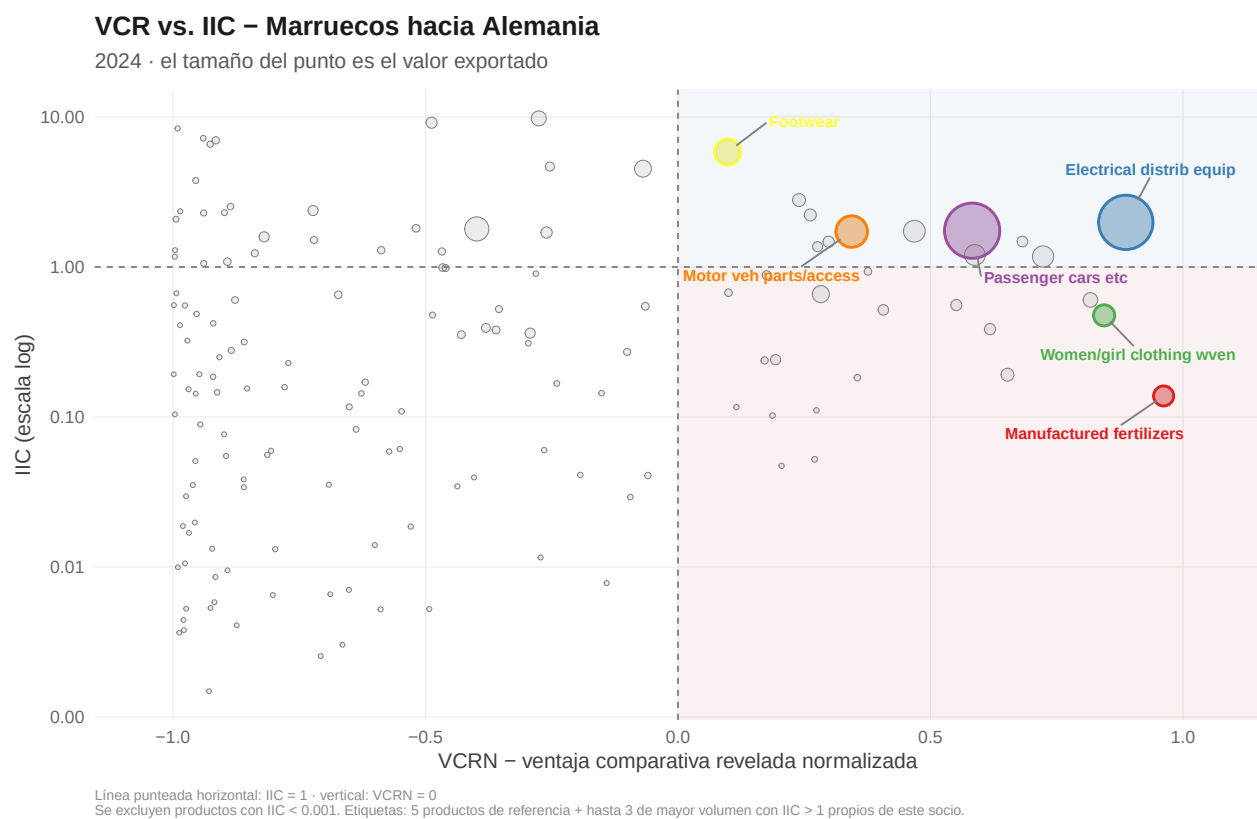


Figure 7: VCR vs. IIC — Marruecos hacia Alemania

```
ggsave(paste0(ruta_graficos, "bubble_DEU.png"), p_deu, width = 10, height = 6.5, dpi = 300, bg = "white")
```

EE.UU.

```
p_usa <- graficar_bubble("USA", iic_min = umbrales_iic[["USA"]])
p_usa
```

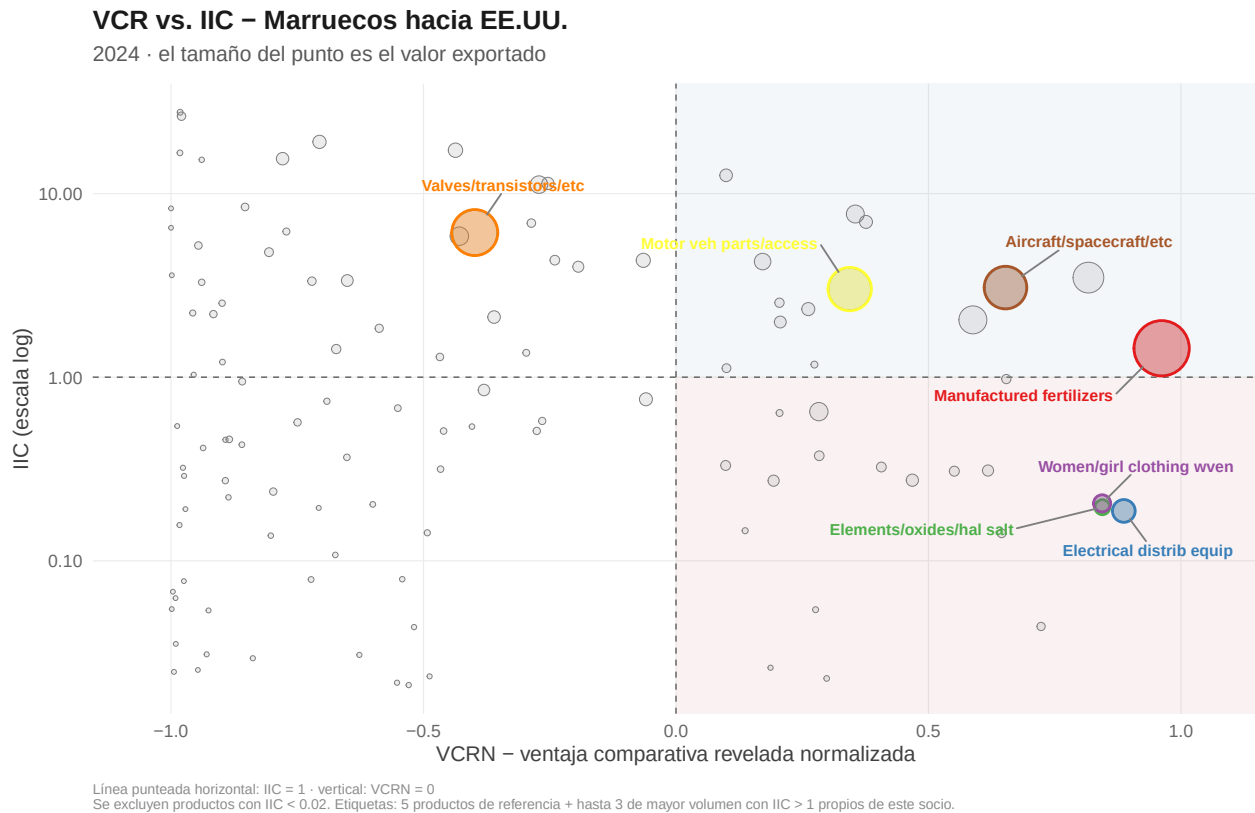


Figure 8: VCR vs. IIC — Marruecos hacia EE.UU.

```
ggsave(paste0(ruta_graficos, "bubble_USA.png"), p_usa, width = 10, height = 6.5, dpi = 300, bg = "white")
```

Brasil

```
p_bra <- graficar_bubble("BRA", iic_min = umbrales_iic[["BRA"]])
p_bra
```

```
ggsave(paste0(ruta_graficos, "bubble_BRA.png"), p_bra, width = 10, height = 6.5, dpi = 300, bg = "white")
```

## Conclusiones

Marruecos muestra una ventaja comparativa muy marcada en fertilizantes (crudos y manufacturados), impulsada por su posición como principal exportador mundial de fosfatos. Sin embargo, este producto no es el que domina la relación bilateral con España pese a ser su principal socio comercial: el comercio con España está traccionado por manufacturas ligadas a cadenas de valor europeas (textiles, equipos eléctricos y autopartes), mientras que los fertilizantes se dirigen mayoritariamente a otros mercados como Brasil e India. A nivel agregado (2 dígitos), la industria automotriz y de equipos eléctricos superan en volumen a los

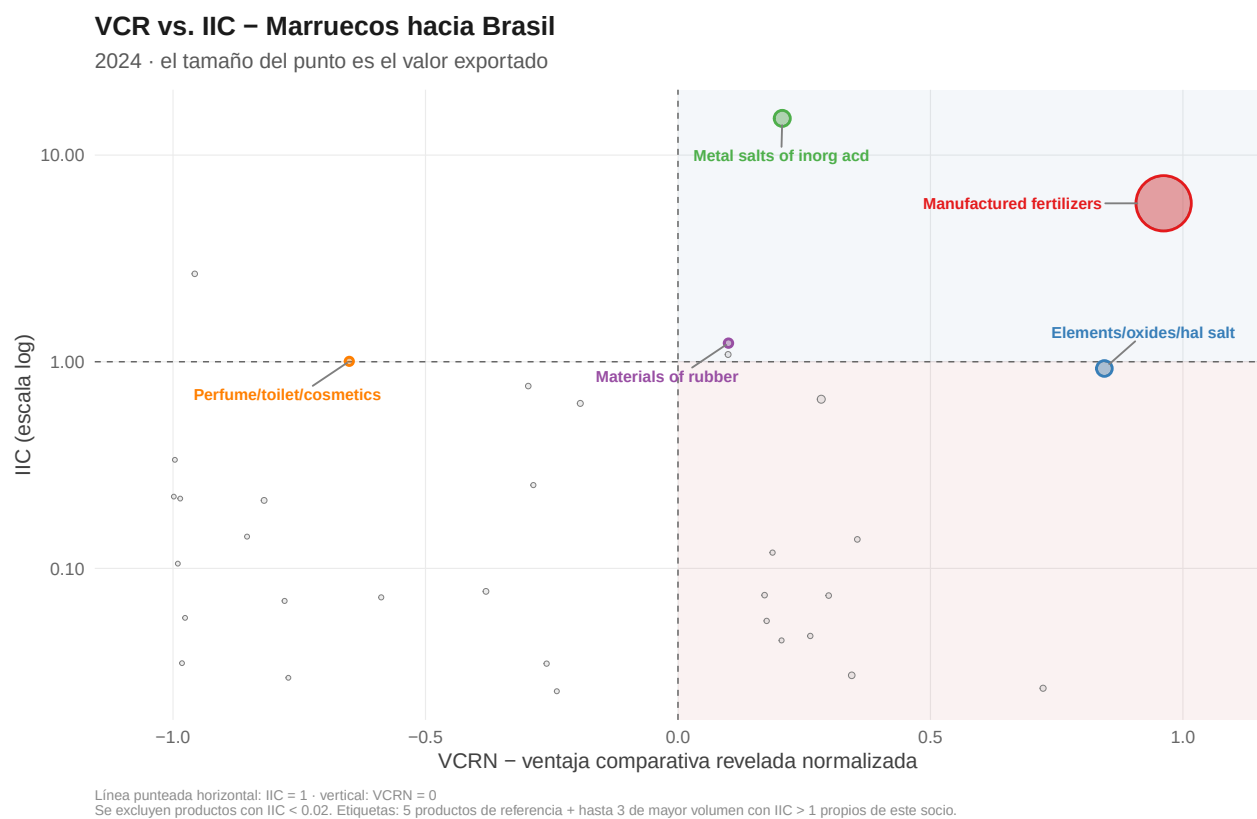


Figure 9: VCR vs. IIC — Marruecos hacia Brasil

fertilizantes, reflejando una integración industrial diversificada que convive con la especialización tradicional en fosfatos.